VU AI HUB · LECTURE NOTES

Computational Thinking

Year 1, September to October 2025

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Solution Strategies Video

How do we approach the problem?

Understand the problem > Choose a strategy > Solve the problem > Success?

If yes: > Test/check

If no: > Choose a different strategy and repeat the steps

Solution Strategies

- Try something (guess and check)
- Go through all the possibilities
  - only suitable if the number of possibilities is limited
- Divide the problem into several sub problems or steps
  - Possible approaches:
    - Simplify
    - Divide
    - Back reasoning
    - Exclude
- Use of formulas/equations
- Discover a structure of pattern
- Make a model
  - visualize the problem and write all the information you can gather from the problem
- Brute force
  - a simple approach to solve problems
  - uses computing power to solve problems with a computer without the use of algorithms or heuristics to speed up the calculation
  - ! only used if no algorithm is known that is faster or more efficient !
- Divide and conquer (D&C)
  1. Divide the problem into a number of small sub problems of the same type and ideally of about the same size (divide)
  2. Solve each sub problem (conquer)
  3. Combine all these solutions into a solution to the original problem (combine)

Mathematical Proof Video

Odd number x Odd number = Odd number

- By definition:
  - n is odd if n = 2k+1
- Proof: m*n is odd
- Suppose:
  - m = 2j+1
  - n = 2k+1
- Then m*n = (2j+1)(2k+1) = 4jk+2j+2k+1 = 2(2jk+j+k)+1
  - all the letters are integers (gehele getallen)
- m*n is of form "2(integer) + 1"
- Hence m*n is odd

Even number - odd number = odd number

- *a* is even and *b* is odd
  - *a* = 2c
  - *b* = 2d + 1
- So:
  - a - b = 2c - (2d+1)

    = 2c - 2d - 1

    = 2c - 2d - 2 + 1

    = 2 (c - d - 1) + 1
    - definition of an odd number
- Hence a - b = odd

3rd of September

4 components of Computational Thinking

- Decomposition
- Patterns
- Abstraction
- Algorithms

Automatic Stock Investment System

Doors: 50 - 1.5% return = 50.75

Pear: 80 - 0.8% return = 80.64

Edison: 40 - 1.8% return = 40.72

D x P = 131.39 → 118.25

D x E = 91.47

P x E = 121.36

- Pear and Edison

HOMA-IR index

- If the glucose levels are given US unit mg/dl then you need to divide them by 18 to get to mmol/l

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Emergency telephone center

60 x 24 = 1440 minutes and calls in a day

1440 / 48 = 30 operators

17th of September

The time complexity is the big O of n (n is your statement)

apparently not always